

Report: Metadata Analysis for ADHD and Sex Prediction

1. Objective

The objective of this analysis was to investigate whether demographic, behavioural and questionnaire-derived metadata could predict:

1. ADHD diagnosis
2. Biological sex

using both machine learning and statistical modelling approaches.

Two complementary modelling strategies were used:

- **LightGBM** for predictive modelling and feature importance.
 - **Logistic Regression** for statistical inference and interpretation of predictor effects.
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2. Dataset

The metadata consisted of demographic, behavioural and clinical questionnaire variables.

Initial dataset:

- Observations: **1,213**
- Variables used for modelling: **24 predictors**

Outcome variables:

- ADHD diagnosis (0 = Control, 1 = ADHD)
 - Sex (0 = Male, 1 = Female)
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3. Data Pre-processing

3.1 Variable classification

Predictors were divided into:

Numerical variables (19)

Included

- MRI age
- Colour vision score

- APQ scores
- SDQ scores
- EHQ score
- Barratt education
- Barratt occupation

Categorical variables (5)

- Enrolment year
 - Study site
 - Ethnicity
 - Race
 - MRI scan location
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3.2 Rare category merging

Before modelling, categorical variables were examined for sparse categories.

Very infrequent categories were merged into an **"Other (99)"** category to reduce instability caused by extremely small group sizes.

This prevents

- unstable coefficient estimates
- inflated standard errors
- convergence problems

during logistic regression.

3.3 Missing value analysis

Several predictors contained missing values.

The largest proportion was

MRI Track Age at Scan

approximately **30% missing**

Smaller proportions occurred in

- Barratt variables
 - questionnaire scores
 - race
 - ethnicity
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3.4 Missing value imputation

A hybrid imputation strategy was used.

Numerical variables

Missing values were imputed using

Iterative Imputation

Characteristics:

- models each variable using the remaining predictors
 - preserves multivariable relationships
 - more informative than mean or median imputation
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Categorical variables

Missing values were replaced using

Mode Imputation

because categorical variables represent discrete classes rather than continuous measurements.

3.5 Encoding

Categorical predictors were converted into dummy variables using one-hot encoding.

After encoding

- Numerical predictors = **19**
- Dummy variables = **19**

Final predictor matrix:

38 predictors

4. Why Barratt variables were modelled numerically

Initially the Barratt education and occupation variables were treated as categorical.

This produced a large number of dummy variables and substantially increased model complexity.

Earlier models using dummy encoding experienced

- convergence problems
- singular matrices
- unstable coefficient estimates

These variables are inherently **ordinal**, representing increasing educational and occupational levels.

Therefore they were retained using their original numeric coding.

Advantages:

- preserved ordering
- reduced dimensionality
- logistic regression converged successfully

This decision was supported by subsequent diagnostic testing.

5. ADHD Prediction using LightGBM

A LightGBM classifier was trained using

- Stratified 5-fold cross-validation

Average performance:

Accuracy $\approx$ **0.786**

Balanced Accuracy $\approx$ **0.728**

Precision $\approx$ **0.819**

Recall $\approx$ **0.883**

F1 $\approx$ **0.850**

ROC-AUC $\approx$ **0.810**

Interpretation

The metadata demonstrated reasonably good predictive performance for ADHD classification.

6. ADHD SHAP Analysis

SHAP values were calculated to explain the LightGBM model.

SHAP provides local and global feature importance.

The summary plot showed that the strongest contributors included questionnaire-derived behavioural measures, particularly SDQ variables.

This analysis guided which predictors were investigated further using logistic regression.

7. Sex Prediction using LightGBM

The same modelling procedure was repeated for Sex.

Cross-validation performance was substantially poorer.

Average ROC-AUC $\approx$ **0.61**

The metadata contained considerably less predictive information for biological sex than for ADHD diagnosis.

8. Logistic Regression

Machine learning models identify predictive relationships but do not provide interpretable effect estimates.

Therefore logistic regression was performed to estimate

- Odds Ratios
- Confidence Intervals
- p-values

for each predictor.

9. Logistic Regression Assumption Checking

Linearity of the logit

The Box–Tidwell procedure was applied to all numerical predictors.

After correcting for multiple comparisons,

no predictor showed evidence of violating the linearity assumption.

Therefore

no polynomial or transformed terms were required.

Multicollinearity

Variance Inflation Factors (VIFs) were calculated.

Largest VIF $\approx$ **2.12**

All VIF values were well below conventional thresholds (5 or 10).

Therefore

multicollinearity was not considered problematic.

10. ADHD Logistic Regression

The final model contained

38 predictors

and converged successfully.

Important findings

Only a small number of predictors showed statistically significant associations after adjustment.

Overall model:

Likelihood Ratio Test

$$\chi^2 = 30.97$$

$$p = 0.784$$

$$\text{McFadden pseudo-}R^2 = 0.0205$$

Interpretation

Although some individual predictors demonstrated significant odds ratios,

the overall metadata model explained only a small proportion of ADHD variability.

11. ADHD Logistic Cross-validation

Cross-validation of logistic regression showed

ROC-AUC $\approx$ **0.49**

Balanced Accuracy $\approx$ **0.49**

This indicates poor predictive performance.

This result contrasts with LightGBM.

Interpretation

Linear logistic regression could not adequately capture the relationships that were successfully modelled by LightGBM.

This suggests that

the metadata likely contains nonlinear relationships and interactions that tree-based models exploit more effectively.

12. Sex Logistic Regression

The Sex logistic model also converged successfully.

Some variables reached nominal statistical significance, including

- SDQ Hyperactivity
- MRI Age
- SDQ Prosocial

However

overall model performance remained weak.

Likelihood Ratio Test

$$\chi^2 = 37.31$$

$$p = 0.501$$

$$\text{McFadden pseudo-}R^2 = 0.0239$$

13. Sex Logistic Cross-validation

Cross-validation results

Balanced Accuracy $\approx$ **0.50**

ROC-AUC $\approx$ **0.52**

indicated essentially random discrimination.

Therefore the metadata did not provide sufficient information for accurate sex classification using logistic regression.

14. Interpretation of LightGBM vs Logistic Regression

The study intentionally used both approaches because they answer different questions.

LightGBM

- focuses on prediction
- captures nonlinear relationships
- identifies influential variables through SHAP

Logistic Regression

- estimates adjusted odds ratios
- provides inferential statistics
- evaluates statistical significance
- offers interpretable coefficients

Together they provide complementary perspectives.

15. Limitations

Several limitations should be acknowledged.

- Only metadata variables were analysed; high-dimensional fMRI connectivity features were not included in this notebook.
- Missing numerical values were handled using a single Iterative Imputation rather than pooled Multiple Imputation.
- Logistic regression assumes linear relationships on the logit scale.
- The ADHD outcome was moderately imbalanced.
- The relatively large number of predictors compared with sample size may reduce statistical power.

16. Conclusion

This analysis demonstrated that metadata alone provides **moderate predictive ability for ADHD when modelled using LightGBM**, achieving a mean ROC-AUC of approximately **0.81**. SHAP analysis identified behavioural questionnaire variables as the most influential predictors.

In contrast, **logistic regression showed limited explanatory power**, with low pseudo- R^2 values and non-significant overall likelihood-ratio tests despite a few individually significant predictors. Cross-validation further indicated poor predictive performance for the logistic models.

For **sex prediction**, both LightGBM and logistic regression exhibited weak discrimination, suggesting that the available metadata contains limited information for distinguishing biological sex.

Overall, the results indicate that **nonlinear machine-learning models are better suited than conventional logistic regression for extracting predictive information from these metadata variables**, while logistic regression remains valuable for interpreting adjusted associations and testing statistical assumptions.

This workflow is appropriate for an MSc project because every preprocessing decision (rare-category merging, iterative imputation, ordinal treatment of Barratt variables, Box–Tidwell testing, VIF assessment, LightGBM, SHAP, and logistic regression) can be clearly justified and defended during a presentation.